

ADITYA KATKAR

Full Stack GenAI Engineer | LLM Systems | Cloud-Native AI Infrastructure

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SUMMARY

Full Stack GenAI Engineer with 2 years of experience building production LLM pipelines, RAG systems, and AI-powered document intelligence platforms for Fortune 500 clients and government bodies. Delivered end-to-end systems using LangChain, FAISS, and vLLM that compressed tender evaluation cycles from 4 weeks of manual work to same-day turnaround, sustained 99.7 percent pipeline reliability across 10,000 daily jobs, and cut cloud spend by 22 percent. Google Cloud Certified Architect and Developer. B.Tech Computer Science, MIT Pune, CGPA 9.2/10.

SKILLS

Generative AI / LLMs: LLM Orchestration, Prompt Engineering, RAG, LangChain, Llama 4 Scout, Gemini 2.5 Flash, BAAI Embeddings, vLLM, Ollama, CUDA, NVIDIA H200

AI / ML: FAISS, BM25 Hybrid Retrieval, EnsembleRetriever, scikit-learn, Neo4j Graph Embeddings, pgvector, QdrantDB, HNSW Indexing

Backend: FastAPI, Flask, Django, Node.js, REST API, GraphQL, Microservices, Celery, RabbitMQ, Kafka

Cloud / DevOps: GCP (GKE, Pub/Sub, Firestore, Cloud Storage, Compute Engine), AWS (Amplify, SES, S3, EC2), Docker, Kubernetes, Terraform, CI/CD

Databases: PostgreSQL, Neo4j, QdrantDB, Redis, MongoDB, Firestore, MSSQL, MySQL

Languages: Python, JavaScript, TypeScript, Rust, SQL

Frontend: React, Next.js, TypeScript, Redux, D3.js, Recharts, Tailwind CSS

WORK EXPERIENCE

Software Engineer | Valiance Solutions, Noida Jan 2025 – Present

- Architected a 6-microservice platform on GKE to fully automate IOCL's vendor tender evaluation. What previously took procurement teams 4 weeks of manual document review now completes on the same day, handling 3 concurrent tenders with 100+ vendor submissions each.
- Built a tiered document ingestion pipeline using Qwen2-VL that automatically processes both digital and scanned PDFs, preserves table structure, and translates Indian-language content. Deployed Llama 4 Scout on vLLM to detect document boundaries across 100+ vendor PDFs per tender, parallelised across 4 workers with InternVL2 as on-premise fallback. Eliminated a manual document preparation step that was blocking the entire evaluation workflow.
- Deployed Qwen 32B to auto-generate Techno-Commercial Comparison reports, with output schemas and prompt templates co-designed with procurement SMEs. Ran air-gapped inside a GCP VPC to satisfy government data-security requirements.
- Built a multi-stage document classification and extraction pipeline using a hybrid BM25 and FAISS retriever (LangChain, BAAI embeddings) that matches vendor submissions against 18 standard procurement templates and extracts 20+ structured compliance data points per vendor including tax IDs, financial turnover, and work-order records. Reduced manual evaluation from weeks to minutes.
- Implemented a fault-tolerant Pub/Sub and Celery pipeline with dead-letter queues and exponential backoff, sustaining 99.7 percent task success across 10,000+ daily AI pipeline jobs.
- Optimised PostgreSQL with composite indexing and a Redis cache-aside layer, reducing P95 latency by 69 percent and monthly GCP spend by 22 percent.
- Streamlined CI/CD on GKE via Bitbucket Pipelines, cutting release cycle from 45 to 8 minutes with zero-downtime deployments.

Python Developer Intern | Markytics.ai, Pune Jan 2024 – Jun 2024

- Built Django REST APIs with RBAC and Celery task queues for 3 production clients, maintaining 85 percent unit test coverage via Pytest.
- Automated a document extraction pipeline processing 500+ documents per day, saving the operations team 12 work-hours every week.
- Shipped real-time shipment tracking using Celery and AWS SES, improving operational visibility by 60 percent and cutting manual interventions by 40 percent.

PROJECTS

IOCL Tender Evaluation Platform Jan 2025 – Present | Indian Oil Corporation Ltd. | GCP

- End-to-end GenAI pipeline: PDF upload triggers a GCS event, tiered OCR and visual segmentation run in parallel via Celery chord, an LLM classifier categorises documents, hybrid BM25 and FAISS RAG matches submissions against procurement templates, structured

compliance data is extracted, and a React dashboard updates in real time. Replaced 4 weeks of manual team effort with a same-day automated result.

- Reduced Firestore read load by 98 percent via Redis mutex batching. Replaced offset-based pagination with keyset pagination to maintain performance at scale.
- Stack: FastAPI, Flask, React, TypeScript, GCP (Pub/Sub, GKE, Firestore), Qwen2-VL, Llama 4 Scout, vLLM, LangChain, FAISS, BM25, Docker, Terraform

Azolla CASPER: Maritime Emissions Compliance SaaS *Jul 2024 – Feb 2025 | Azolla, Singapore | AWS*

- Built an ML-powered EU ETS carbon penalty exposure calculator and FuelEU Maritime emissions engine using scikit-learn. Produced 30-day penalty forecasts with average error under 50,000 EUR, validated by maritime domain experts.
- Launched Pool Dive, a compliance credit marketplace using algorithmic vessel-matching on emissions balance, GHG intensity delta, and trade route, with surplus modelled as an internal green credit currency.
- Configured RBAC dashboards for fleet managers, compliance officers, and vessel operators with automated nightly SES threshold alerts for carbon allowance and fuel emissions breaches.
- Stack: React, Django, AWS (Amplify, SES), MSSQL, Celery, Redis, scikit-learn

KM-Tech-Int Digital Forensics Platform *Jul 2024 – Present | Malaysian Government | GCP*

- Integrated Gemini 2.5 Flash to extract structured forensic identifiers (IMSI, IMEI, MAC, IP) from VoIP logs, CDRs, and court PDFs in a single-pass LLM extraction with no chunking required, ingested into Neo4j with full relational context.
- Reduced case analysis time by 10x over manual SQL workflows by enabling multi-hop link analysis through Neo4j Cypher graph traversals.
- Delivered a D3.js force-directed graph in React with live node expansion, path highlighting, and court-admissible SVG export.
- Enabled case precedent matching via pgvector semantic search over document embeddings, surfacing similar past investigations in under 30 seconds.
- Stack: React, D3.js, Django, Neo4j, Gemini 2.5 Flash, pgvector, PostgreSQL, GCP, Docker

CERTIFICATIONS

- Google Professional Cloud Developer, Dec 2024 [View Credential](#)
- Google Professional Cloud Architect, Jan 2025 [View Credential](#)

EDUCATION

B.Tech Computer Science | MIT Pune *Oct 2020 – May 2024*

CGPA 9.2 / 10 | Top 10% of graduating class